

MUSTHOFA JOKO ANGGORO

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RESEARCH INTERESTS

Edge AI systems, resource-aware computing in edge-fog-cloud continuum, sustainable computing, energy-efficient systems, distributed systems architectures, hardware-software co-design

EDUCATION

University of Indonesia (UI) Jakarta, Indonesia
Bachelor of Computer Science, Information Systems *Expected Graduation: Aug 2026*
• **GPA: 3.84/4.00** (8th Semester, Dean's List)

- **Thesis:** Implementation of Morris Mano Computer Architecture with Enhancements on Xilinx Spartan-3 FPGA (X3700AN) using VHDL — demonstrating hardware design, digital logic optimization, and system-level architecture implementation
- **Relevant Coursework:** Computer Architecture, Operating Systems, Data Structures & Algorithms, Introduction to AI & Data Science, Statistics & Probability, Data Communication Networks, Database Systems

Nanyang Technological University (NTU) & ITS Singapore & Surabaya
INSPIRASI-NTU Summer Program 2025 (LPDP Scholarship Awardee) *Jan 2025 - Feb 2025*

- Conducted sustainability-focused research on smart port automation and green technology adoption in distributed maritime systems, analyzing IoT sensor networks, automation architectures, and energy-efficient deployments
- Produced comprehensive technical report proposing tiered port automation framework emphasizing renewable energy utilization and carbon emission reduction — directly aligned with sustainability objectives in edge computing systems
- Collaborated with international research teams on applying systems thinking to environmental challenges

SYSTEMS & RESEARCH EXPERIENCE

Software Engineer Intern Traveloka (Travel Technology Company)
Systems Engineering & Backend Development *Aug 2025 - Present*

- Engineered production-grade distributed systems including credential management with AES-GCM encryption, automated rotation snapshots, and granular audit logging using MongoDB for high-throughput operations
- Developed scalable automation services with configurable scrapers and failover mechanisms ensuring 99.9% availability, processing thousands of daily transactions with auto-recovery logic
- Enhanced CI/CD pipelines with automated API testing (Karate framework) and auto-rollback logic to maintain system reliability and prevent production defects in microservice architecture
- Built real-time data synchronization services and optimized structured logging for BigQuery analytics, demonstrating proficiency in distributed data flow and performance optimization
- **Systems Tools:** Java, Spring Boot, MongoDB, Google Cloud Platform, Docker, Kubernetes, BigQuery

Full Stack Developer PT Magna Solusi Indonesia
Enterprise Systems Development *Jan 2025 - Jun 2025*

- Developed full-stack enterprise modules with focus on database performance optimization, query efficiency, and backup strategies for high-reliability production systems
- Built RESTful APIs supporting dynamic data flow in distributed web architecture with Spring Boot microservices
- **Systems Tools:** Spring Boot, Angular, PostgreSQL, Apache HTTP Server, Docker, Kubernetes

Lead Engineer & Full Stack Developer Event Organizer Enterprise System (Course Project)
System Architecture & Implementation *Feb 2025 - May 2025*

- Led engineering team in designing and implementing enterprise-scale system with focus on clean architecture, secure authentication (JWT), and containerized deployment demonstrating leadership in systems development
- Architected full-stack solution with PostgreSQL optimization, RESTful API design, and CI/CD integration
- **Systems Tools:** Spring Boot, Vue.js, PostgreSQL, Docker, TypeScript

TECHNICAL SKILLS

Systems Programming: Java, Python, C (VHDL for hardware), TypeScript, Go
Hardware & Architecture: VHDL, FPGA development (Xilinx Spartan-3), digital logic design, computer architecture implementation
Systems & Frameworks: Spring Boot, Django, microservices architecture, RESTful API design
Databases & Storage: PostgreSQL, MongoDB, manual SQL optimization, query performance tuning
DevOps & Infrastructure: Docker, Kubernetes, CI/CD pipelines, Google Cloud Platform
AI/ML Fundamentals: Scikit-learn, supervised/unsupervised learning, data preprocessing, model evaluation
Operating Systems: Linux, Unix, Windows — familiar with OS internals, process management, memory management

LEADERSHIP & SERVICE

Vice Person in Charge of UI/UX <i>Design System & Cross-functional Collaboration</i>	COMPFEST (Annual Tech Festival, UI) <i>Jan 2023 - Dec 2023</i>
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- Led 11-member design team in creating scalable design system and managing 5 development sprints for compfest.id website, demonstrating project management and cross-functional collaboration skills
- Coordinated with software and automation engineers to ensure design-engineering alignment, bridging technical and design perspectives in system development

Mentor, Programming Foundations 0 — University of Indonesia (2024)
Relevant Projects: Machine learning predictive modeling (imbalanced data, Scikit-learn), database-driven web systems (Django, PostgreSQL with triggers/stored procedures), microservices architecture (Spring Boot)